

Problem 5 Solution – Cracked Section Analysis

Section properties:

$$A_g = 320,000 \text{ mm}^4$$

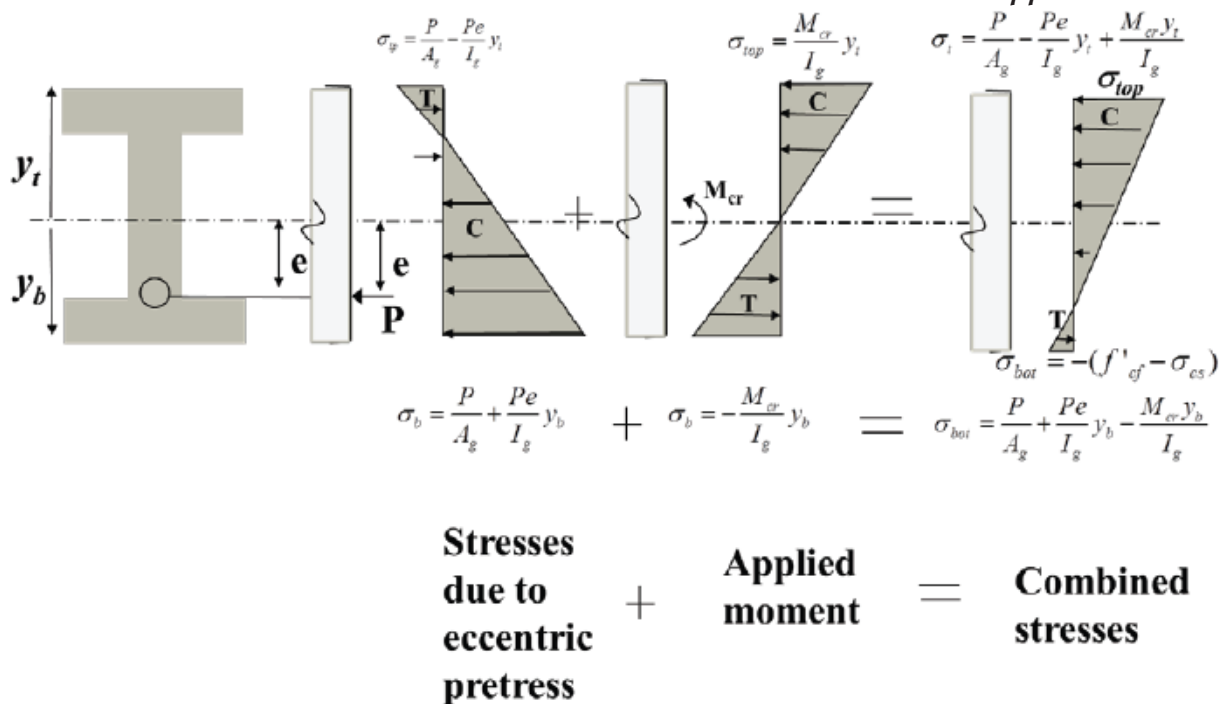
$$P = A_p \sigma_{pe} = 1000 \times 1325 = 1325 \times 10^3 \text{ N}$$

$$I_g = 3.857 \times 10^{10} \text{ mm}^4$$

$$e = 350 \text{ mm}$$

$$y_b = 500 \text{ mm}$$

1. Demonstrate that the beam is cracked under the applied



Stress at the bottom fibre when beam is cracking is given by:

At the cracking moment, σ_{bot} will reach cracking tensile stress of concrete,

$0.6\sqrt{f'_c} - \sigma_{cs}$. Stress at the bottom fibre when beam is cracking is given by:

$$\sigma_b = -\left(0.6\sqrt{f'_c} - \sigma_{cs}\right) = \frac{P}{A_g} + \frac{Pe y_b}{I_g} - \frac{M_{cr} y_b}{I_g}$$

Where:

$$0.6\sqrt{f'_c} = 0.6\sqrt{32} = 3.39$$

$$\sigma_{cs} = E_s \varepsilon_{cs}^* (2.5p_w - 0.8p_{cw}) / (1 + 50p_w) \quad \text{Chapter 3 AS3600}$$

$$d = (d_p + d_s) / 2 = (850 + 963) / 2 = 907 \text{ mm}$$

$$p_w = (A_{st} + A_p) / (b_w d) = (3600 + 1000) / (200 * 907) = 0.025$$

$$p_{cw} = A_{sc} / b_w d = 1800 / (200 * 907) = 0.01$$

$$\varepsilon_{cs}^* = 556 \times 10^{-6} \quad \text{Tb 3.1.72 AS3600, where}$$

$$t_h = 2A_g / u_e = (2 * 320,000) / 4000 = 160 \text{ mm}$$

$$\sigma_{cs} = 200,000 \varepsilon_{cs}^* (2.5p_w - 0.8p_{cw}) / (1 + 50p_w) = 2.69$$

$$\sigma_{bot} \leq -(0.6\sqrt{f'_c} - \sigma_{cs}) \quad (\text{remember that -'ve represents tension})$$

$$\sigma_b = -(0.6\sqrt{f'_c} - \sigma_{cs}) = \frac{P}{A_g} + \frac{Pey_b}{I_g} - \frac{M_{cr} \cdot y_b}{I_g}$$

$$\frac{P}{A_g} = \frac{1325 \times 10^3}{320,000} = 4.14 \text{ MPa}$$

$$\frac{Pey_b}{I_g} = \frac{1325 \times 10^3 \times 350 \times 500}{3.857 \times 10^{10}} = 6.01 \text{ MPa}$$

$$\sigma_{bot} = -(3.39 - 2.69) = 4.14 + 6.01 - \frac{M_{cr} \times 500}{3.857 \times 10^{10}}$$

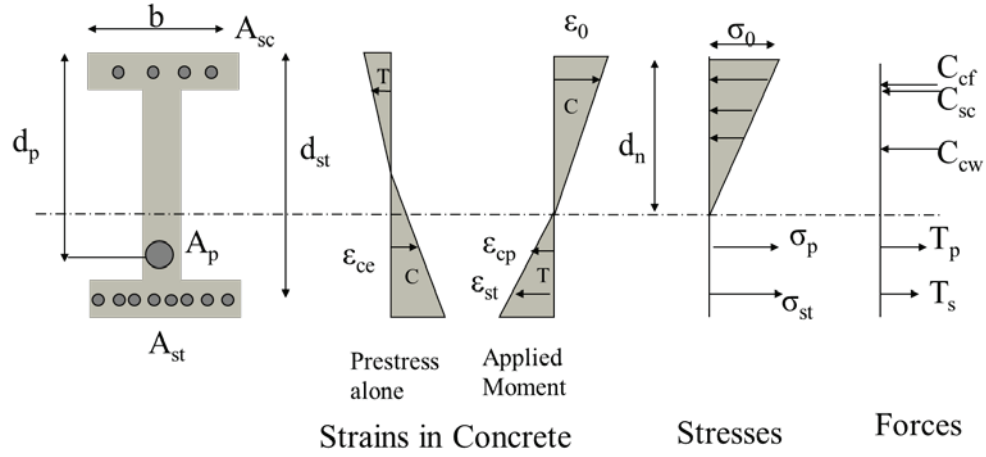
$$M_{cr} = 837 \text{ kNm}$$

$$\therefore \text{cracking moment} = 837 \text{ kNm}$$

$$\text{service bending moment} = \frac{\omega L^2}{8} = \frac{50 \times 16^2}{8} = 1600 \text{ kNm}$$

Since applied bending moment, $1615 \text{ kNm} > M_{cr} (837 \text{ kNm})$, the beam is cracked under the service load.

2. Cracked section analysis



Section properties:

$$d_p = 1000 - 150 = 850 \text{ mm}$$

$$d_{st} = 1000 - 25 - \frac{24}{2} = 963 \text{ mm}$$

$$d_{sc} = 25 + \frac{24}{2} = 37 \text{ mm}$$

$$E_c = 0.043\sqrt{32} \times 2400^{1.5} = 28600 \text{ MPa}$$

$$E_{cp} = 0.043\sqrt{f_{cp}} p^{1.5} \text{ (@ transfer)} = 0.043\sqrt{25} \times 2400^{1.5} = 25279 \text{ MPa}$$

$$E_p = 190,000 \text{ MPa}$$

$$A_p = 1000 \text{ mm}^2$$

ϵ_{pe} = strain due to prestress applied on cable

$$\epsilon_{pe} = \frac{P}{A_p E_p} = \frac{1325 \times 10^3}{1000 \times 190,000} = 0.00697$$

ϵ_{ce} = concrete compressive strain at the level of tendon

$$\epsilon_{ce} = \frac{1}{E_c} \left[\frac{P}{A} + \frac{Pe}{y} \right]$$

$y = 350 \text{ mm}$ (we are calculating stress at the level of tendon)

$$\begin{aligned}\varepsilon_{ce} &= \frac{1}{28600} \left[\frac{1325 \times 10^3}{320,000} + \frac{1325 \times 10^3 \times 350 \times 350}{3.857 \times 10^{10}} \right] \\ &= \frac{1}{28600} [4.14 + 4.21] = 0.0003\end{aligned}$$

Calculation forces in section $C_c = C_f + C_w$:

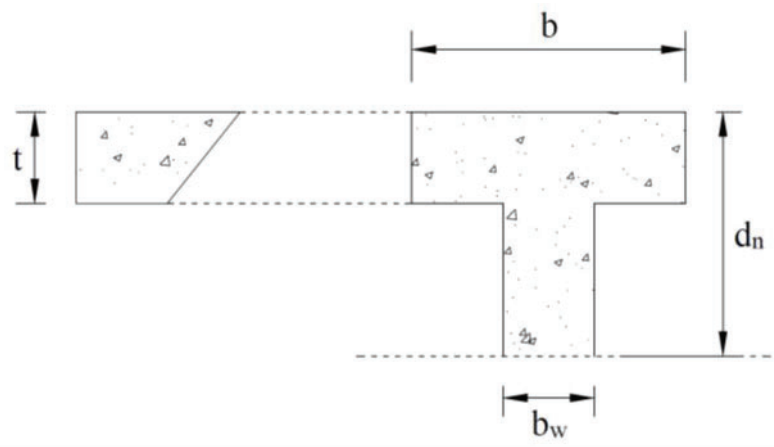


Figure S4.2, Calculation of forces, copyright RMIT University

Force in flange:

$$\begin{aligned}C_f &= 0.5 \times (\sigma_o + \sigma_t) (b - b_w) t \\ &= 0.5 \times \left[E_c \varepsilon_o + E_c \varepsilon_o \times \frac{(d_n - t)}{d_n} \right] (b - b_w) t \\ &= 0.5 \times \left[28600 \varepsilon_o + 28600 \varepsilon_o \times \frac{(d_n - 150)}{d_n} \right] (600 - 200) \times 150 \\ &= 0.5 \times 28600 \times \varepsilon_o \times 400 \times 150 \left[1 + \frac{(d_n - 150)}{d_n} \right] \\ &= 8.58 \times 10^8 \times \varepsilon_o \left[\frac{2d_n - 150}{d_n} \right]\end{aligned}$$

Force in web:

$$\begin{aligned} C_w &= 0.5\sigma_o b_w d_n = 0.5 \times E_c \times \varepsilon_o \times b_w \times d_n \\ &= 0.5 \times 28600 \times \varepsilon_o \times 200 \times d_n \\ &= 2.86 \times 10^6 \times \varepsilon_o \times d_n \end{aligned}$$

Total force in flange and web in concrete:

$$C_c = 8.58 \times 10^8 \times \varepsilon_o \left[\frac{2d_n - 150}{d_n} \right] + 2.86 \times 10^6 \times \varepsilon_o \times d_n$$

Force in top reinforcement:

$$\begin{aligned} C_s &= E_s A_{sc} \times \frac{\varepsilon_o}{d_n} (d_n - d_{sc}) = 200,000 \times 1800 \times \frac{\varepsilon_o}{d_n} (d_n - 37) \\ &= 3.6 \times 10^8 \times \varepsilon_o \left(\frac{d_n - 37}{d_n} \right) \end{aligned}$$

Force in bottom reinforcement:

$$\begin{aligned} T_s &= E_s A_{st} \varepsilon_o \times \left(\frac{d_{st} - d_n}{d_n} \right) = 200,000 \times 3600 \times \varepsilon_o \left(\frac{963 - d_n}{d_n} \right) \\ &= 7.2 \times 10^8 \times \varepsilon_o \left(\frac{963 - d_n}{d_n} \right) \end{aligned}$$

Force in cables:

$$\begin{aligned} T_p &= E_p A_p \times (\varepsilon_{ce} + \varepsilon_{pe} + \varepsilon_{cp}) \\ &= 190,000 \times 1000 \times \left(0.0003 + 0.00697 + \frac{\varepsilon_o (d_p - d_n)}{d_n} \right) \\ &= 1.9 \times 10^{10} \times \left[\frac{\varepsilon_o}{d_n} (850 - d_n) + 0.0073 \right] \end{aligned}$$

Consider force equilibrium:

$$\overbrace{(C_f + C_w)}^{C_c} + C_s = T_s + T_p$$

We have all the forces in the above equation as a function of ε_o and d_n . For a manual solution, usual procedure is assuming a value for ' d_n ' and calculating ' ε_o ' using force equilibrium. This can be easily performed using a spread sheet. Finally check whether the neutral axis is in the web.

Using the equation derived in Week 4 lecture notes.

$$\begin{aligned}
 \varepsilon_0 &= \frac{E_p A_p d_n (\varepsilon_{ce} + \varepsilon_{pe})}{\left[0.5 E_c b_w d_n^2 + E_c (d - 0.5t)(b - b_w)t + E_s A_{sc} (d_n - d_{sc}) - E_s A_{st} (d_{st} - d_n) - E_p A_p (d_p - d_n) \right]} \\
 &= \frac{190,000 \times 1000 \times d_n (0.0003 + 0.00697)}{\left[0.5 \times 28600 \times 200 \times d_n^2 + 28600 (d_n - 0.5 \times 150)(600 - 200) \times 150 + 190,000 \times 1800 (d_n - 37) - 200,000 (850 - d_n) \right]} \\
 &= \frac{1387000 d_n}{2,860,000 d_n^2 + 1.716 \times 10^9 (d_n - 75) + 3.42 \times 10^8 (d_n - 37) - 7.2 \times 10^8 (963 - d_n) - 1.9 \times 10^8 (850 - d_n)} \\
 &= \frac{d_n}{2.06 d_n^2 + 1237 (d_n - 75) + 247 (d_n - 37) - 519 (963 - d_n) - 137 (850 - d_n)} \\
 &= \frac{d_n}{2.06 d_n^2 + 2140 d_n - 718161}
 \end{aligned}$$

Assume $d_n = 500$ (neutral axis in the web)

$$\varepsilon_0 = 0.00058$$

$$\sigma_0 = E_c \varepsilon_0 = 28600 \times 0.00058 = 16.6 \text{ MPa}$$

$$\begin{aligned}
 C_c &= 8.58 \times 10^8 \times \varepsilon_0 \left[\frac{2d_n - 150}{d_n} \right] + 2.86 \times 10^6 \times \varepsilon_0 \times d_n = 8.58 \times 10^8 \times 0.00058 \times \left[\frac{2 \times 500 - 150}{500} \right] + 2.86 \times 10^6 \times 0.00058 \\
 &= 1,675,388 \text{ N}
 \end{aligned}$$

$$C_s = 3.6 \times 10^8 \times \varepsilon_0 \left[\frac{d_n - 37}{d_n} \right] = 3.6 \times 10^8 \times 0.00058 \times \left[\frac{500 - 37}{500} \right] = 193,349 \text{ N}$$

$$T_s = 7.2 \times 10^8 \times \varepsilon_0 \left[\frac{963 - d_n}{d_n} \right] = 7.2 \times 10^8 \times 0.00058 \times \left[\frac{963 - 500}{500} \right] = 386,698 \text{ N}$$

$$T_p = 1.9 \times 10^8 \left[\frac{\varepsilon_0}{d_n} \times (850 - d_n) + 0.0073 \right] = 1.9 \times 10^8 \times \left[\frac{0.00058}{500} \times (850 - 500) + 0.0073 \right] = 1,464,140 \text{ N}$$

To check whether the neutral axis is in the web, moment equilibrium is considered.

Take moment about top fibre:

$$\begin{aligned}
 M &= T_s d_{st} + T_p d_p - C_w d_n / 3 - C_f d_f - C_s d_{sc} \\
 &= \underbrace{386698 \times 963}_{T_s d_{st}} + 1464140 \times 850 - (2.86 \times 10^6 \times 0.00058 \times 500 \times 500 / 3) - 8.58 \times 10^8 \times 0.00064 \left[\frac{2 \times 500 - 150}{500} \right] \\
 d_f &= \frac{t(3d_n - 2t)}{(6d_n - 3t)} = \frac{150(3 \times 500 - 2 \times 150)}{(6 \times 500 - 3 \times 150)} = 71 \text{ mm} \\
 \therefore M &= 1,450,305,144 \text{ Nmm} \\
 M_{\text{applied}} &= 1450 \text{ kNm}
 \end{aligned}$$

Applied BM due to factored service load is 1600 kNm.

When we assume $d_n = 500 \text{ mm}$ and our calculated $M_{\text{applied}} = 1450 \text{ kNm}$ our assumption is quite close. If it had been significantly different you may try a different value of ' d_n ' until calculated BM = 1600 kNm. This can easily be performed by using a spreadsheet.